Project Executable Files

Date	04 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
OptiCrop/
│
├── app.py
├── requirements.txt
├── README.md
│
├── model/
│   ├── model.pkl
│   └── label_encoder.pkl
│
├── dataset/
│   └── Crop_recommendation.csv
│
├── notebooks/
│   └── Model_Training.ipynb
│
├── static/
│   ├── css/
│   │   ├── style.css
│   │   └── animations.css
│   │
│   ├── js/
│   │   └── script.js
│   │
│   └── images/
│
├── templates/
│   ├── home.html
│   ├── about.html
│   ├── predict.html
│   ├── result.html
│   ├── contact.html
│   ├── navbar.html
│   └── footer.html
│
└── utils/
    ├── preprocessing.py
    └── predictor.py
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not yet deployed — currently runs locally using Flask.
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Server
Repository Link	OptiCrop: Smart Agricultural Production Optimization Engine
Demo Video Link	Submitted along with the project files.

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites

- Python 3.10 or above
- VS Code
- Flask
- Required Python libraries

Steps

1. Download or clone the project.
2. Open the project folder in VS Code.
3. Install dependencies:
   pip install -r requirements.txt
4. Run the application:
   python app.py
5. Open the browser and visit:
   http://127.0.0.1:5000
6. Enter soil and weather values to get crop recommendations.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction depends on dataset quality	Use accurate dataset
2	Internet not required, but no live weather data	Future enhancement
3	Supports limited crop dataset	More crops can be added later